

Graduate Seminar in Differential Geometry: Differentiable Stack Cohomology

C. Blohmann, A. Kraasch-Tarnowsky

Updated information on the seminar and the meeting can be found at <https://guests.mpim-bonn.mpg.de/tarnowsky/seminar26/>.

Topic of the seminar: Higher differential geometry is a field of study at the intersection of differential geometry and category theory. Differential geometry generally concerns itself with studying manifolds, where category theory can be used to introduce e.g. notions of symmetry, leading to a rich field of study of these so-called higher structures. The seminar talks can be grouped into 3 different sections:

- A. Lie groups and equivariant cohomology
- B. Lie groupoids
- C. Differentiable stacks.

Target group: Master's students interested in differential geometry. Ambitious Bachelor students interested in differential geometry may also join.

Prerequisites: Some knowledge on manifolds (i.e. definition of smoothness, tangent spaces, vector fields, metrics etc.) and some very basic knowledge on categories (i.e. definition of a category, functor, natural transformation, etc.)

About the talks:

- The format of the seminar will be a series of **35-40 minute** blackboard talks. Expect around 5 minutes of additional questions.
- You will be graded not only for the mathematical correctness of your talk and your understanding of the material, but also for the quality of your presentation (clarity of explanations, engaging, etc.).
- The talks have a difficulty score of (*) to (***). We will take the difficulty into consideration when grading, but an easier talk does not mean that you don't have a chance at a top grade.
- It is **strongly recommended to practice** your talk (at best in a classroom) before presenting, especially as the **time limit will be strict**.

- Please schedule a preliminary meeting with us before your talk. The meeting should take place about a week before the talk.
- You have to prepare written notes for your talk and bring them with you for the preliminary meeting (likely 1-2 pages double-sided, but it strongly depends on your handwriting).

About the seminar:

- The seminar takes place each Wednesday, 14 - 16 h, in room 1.008 at the math institute, with the exception of 6.5. (both instructors will be travelling), 20.5. (Dies Academicus) and 27.5. (Pentecost)
- The registration for the seminar consists of registering in BASIS in addition to the registration with the organisers.
- For the distribution of talks, we kindly ask you to send an email to the organisers. The email should contain at least 3 topics you can see yourself presenting. These topics should come from at least 2 of the 3 areas (i.e. not all from the same one).
- We expect you to **attend the seminar every week**. Please contact us in case you are sick or need to be absent for other reasons. You may not be absent for more than two appointments in total. For each absence, there may be replacement coursework on the missed topic.
- The schedule is preliminary (it might be necessary to shift the talks by one week during the semester).

Schedule: The schedule of the talks is as follows:

- A1. Lie groups and Lie algebras (15.4.2026)
- A2. Lie group actions on manifolds (15.4.2026)
- A3. Quotients of Lie group actions (22.4.2026)
- A4. Equivariant cohomology (22.4.2026)
- A5. Properties of equivariant cohomology (29.4.2026)
- A6. The Weil algebra (29.4.2026)
- A7. Models for equivariant cohomology (13.5.2026)
- A8. The equivariant localisation formula (13.5.2026)

- B1. Lie groupoids and Lie algebroids (3.6.2026)
- B2. Lie groupoid actions and principal bundles (10.6.2026)
- B3. Morita bibundles (10.6.2026)
- B4. Simplicial manifolds and the Bott-Shulman-Stasheff complex (17.6.2026)
- B5. The nerve construction (17.6.2026)
- C1. Pre-stacks over $\mathbf{Mfld}$ (24.6.2026)
- C2. Differentiable stacks (1.7.2026)
- C3. Differentiable stacks vs. Lie groupoids (1.7.2026)
- C4. Sheaves on differentiable stacks (8.7.2026)
- C5. Hypercohomology of differentiable stacks (8.7.2026)
- C6. Cohomology of differentiable stacks via atlases (15.7.2026)

On 22.7. there will be a talk on infinitesimal models for differentiable stack cohomology by the organisers.

Programme:

A1. Lie groups and Lie algebras(*):

- Define Lie groups and Lie algebras and their homomorphisms and give examples
- Explain differentiation, define the exponential map
- Give some of the statements on the correspondence between Lie groups and Lie algebras without too much detail
- Source recommendations: Chapters 7, 8, 20 of [13]

A2. Lie group actions on manifolds(*):

- Define Lie group actions and give examples including adjoint and coadjoint action
- Define equivariant maps, fundamental vector fields
- Define a Lie algebra action and explain how to derive a Lie group action
- Source recommendations: Chapters 7, 20 in [13], Sections 1.1-1.3 and 11.1 in [17]

A3. Quotients of Lie group actions(**):

- Define proper/free/transitive actions, orbit spaces, fixed points
- Recall the definition of principal bundles
- State Theorem 21.10 of [13] and comment on the proof
- Source recommendations: Chapters 7, 21 in [13], Sections 1.1-1.3 and 11.1 in [17]

A4. Equivariant cohomology(**):

- Explain the problem of defining “the cohomology of orbit spaces”, introduce the homotopy quotient, define equivariant cohomology
- Give examples
- Sketch that for free actions, the intuition and the definition coincide
- Source recommendations: Chapters 4, 9 of [17]

A5. Properties of equivariant cohomology(**):

- Describe properties of equivariant cohomology (e.g. equivariant morphisms induce morphisms in cohomology, $H(BG)$ -algebra structure)
- State the equivariant de Rham theorem on the Cartan model (Theorem 21.6) without explaining the differential, use it to compute the cohomology of a group acting on a point
- Source recommendations: Chapters 4, 9, 21 of [17]

A6. The Weil algebra(***):

- Define the notion of $\mathcal{G}^*$ -algebra, describe the $\mathcal{G}^*$ -algebra structure of $\Omega(M)$ for a manifold with a $\mathcal{G}$ -action
- Translate the conditions of contractibility and free action of $E\mathcal{G}$ to an imagined $\Omega(E\mathcal{G})$, then define the Weil algebra as the Koszul algebra of $\mathfrak{g}^*$ and its $\mathcal{G}^*$ -algebra structure
- Describe the change of coordinates to horizontal generators of $\mathcal{W}(\mathfrak{g})$ including the new equations for the differential etc.
- Source recommendations: Chapters 19-21 of [17], Sections 2.3, 3.1, 3.2, 4.1, 4.2 of [10]

A7. Models for equivariant cohomology(***):

- Motivate why the basic subcomplex of $\Omega(M) \otimes \mathcal{W}(\mathfrak{g})$ should describe a model for equivariant cohomology, state the equivariant de Rham theorem

- Describe the Mathai-Quillen isomorphism and how it transforms the G^* -algebra structure and differential
- Describe how the differential simplifies in the Cartan model
- Source recommendations: Chapters 19-21 of [17], Sections 2.3, 3.1, 3.2, 4.1, 4.2 in [10]

A8. The equivariant localisation formula(**):

- State the equivariant localisation formula for a circle action, explain the setting and the occurring coefficients
- Give a rough intuition why some kind of equivariant localisation formula for a circle action is to be expected
- Compute the area of a sphere using the equivariant localisation formula
- Source recommendations: Chapter 27-30 in [17]

B1. Lie groupoids and Lie algebroids(*):

- Motivate the notion of groupoids, define Lie groupoids, give examples and in particular include the action groupoid of a Lie group action
- Define Lie algebroids and describe the differentiation from Lie groupoids to Lie algebroids
- Comment shortly on local to global results on Lie groupoids and Lie algebroids
- Source recommendations: Sections 1-4 in [5], Sections 1, 3, 6, 7 and 9 in [14]

B2. Lie groupoids actions and principal bundles(*):

- Define the action of a Lie groupoid, give examples including the canonical action on the manifolds of the groupoid itself
- Define the notion of a Lie groupoid principal bundle, give examples
- Define the notion of pull-backs and trivial principal bundles, relate the definition to “local triviality”
- Source recommendations: Section 5 in [5], Section 5.7 in [15], Section 5 in [14]

B3. Morita bibundles(**):

- Define the notion of Morita bibundles and the category structure they induce on Lie groupoids (i.e. composition of bibundles)

- Define the notion of Morita equivalence and give examples
- Compare to the notion of a Morita morphism
- Source recommendations: Section 2.2 and 2.3 in [3], Sections 4.3-4.6 in [7]

B4. Simplicial manifolds and the Bott-Shulman-Stasheff complex(***):

- Define a simplicial set, give examples, draw pictures
- Define a simplicial object in the category of manifolds
- Review the definition of the Moore complex of a simplicial abelian group, define the Bott–Shulman–Stasheff bicomplex and its differentials
- Define geometric realisation, state the Bott–Shulman–Stasheff theorem on the cohomology of the realisation of a simplicial manifold
- Source recommendations: Sections 2.4,2.5 in [6], Section I.1, III.2 in [9]
- Recommended prerequisites: Know simplicial sets

B5. The nerve construction(**):

- Define the simplicial structure of the nerve of a Lie groupoid
- Prove that equivalent topological categories have homotopic nerve realisation (Proposition 2.1 in [16]) and use this to show that the nerve realisation of $\mathcal{G} \circ \mathcal{G}$ can be used as $E\mathcal{G}$
- sketch how the homotopy quotient of equivariant cohomology arises as the nerve realisation of the action groupoid
- Source recommendations: Section 2, 3 in [16]
- Recommended prerequisites: Know simplicial sets

C1. Pre-stacks over $\mathbf{Mfld}(*):$

- Consider the definition of a pre-stack over $\mathbf{Mfld}$, give some nice examples
- Compare to the formulation of pre-stacks in [11] (i.e. ignoring the descent conditions)
- define morphisms and 2-morphisms of stacks (i.e. 2-category structure), show that $\mathbf{Mfld}$ is a subcategory of the category of stacks, remark on the classifying stack $B\mathcal{G}$
- Source recommendations: Sections 1.1, 1.3 in [8], Section 2.1 in [2], Section 1 in [11]

C2. Differentiable stacks(**):

- Introduce the descent conditions of a stack over $\mathbf{Mfld}$, sketch it in examples
- Spell out the fibre product of stacks and state its universal property
- Describe when a morphism is representable and when a morphism is an epimorphism, then combine into a description of a representable epimorphism
- Source recommendations: Sections 2.1-2.3 in [2], Sections 1.3, 2.1 in [8], Section 2 in [11]

C3. Differentiable stacks vs. Lie groupoids(**):

- Define the Lie groupoid associated to an atlas, define the stack associated to a Lie groupoid
- Sketch the proof of the equivalence of 2-categories between differentiable stacks and Lie groupoids
- Source recommendations: Sections 2.4-2.6 in [2], Section 2.2 in [8], Section 2 in [3]

C4. Sheaves on differentiable stacks(***):

- Expect no prior knowledge of sheaf theory
- Define a sheaf on a topological space and a sheaf on a site, define the necessary notions for later
- Give both definitions of a sheaf on a stack and relate to both notions, give the trivial and locally constant sheaf as an example
- Define cartesian sheaves, show that the locally constant sheaf is cartesian and give an example of a non-cartesian sheaf
- Source recommendations: Section 3.1 of [1], Section 4 of [11], Chapter II in [12], Chapter I in [4]
- Recommended prerequisites: Know basics about sheaves (stalks, sheafification, push-forward)

C5. Hypercohomology of differentiable stacks(***):

- Review the notion of a Cartesian sheaf and explain the alternate description of a cartesian sheaf as a sheaf on an atlas
- Define the global sections functor following [2], note that for cartesian sheaves it is the equaliser
- Sketch the concept of a right derived functor and motivate why to think of cohomology as the right derived functor of the global sections functor, define hypercohomology groups with values in a sheaf of a differentiable stack

- Source recommendations: Section 3.1 of [1], Section 4 of [11]
- Recommended prerequisites: Know what a right derived functor is, know what Čech cohomology is

C6. Cohomology of differentiable stacks via atlases(***):

- Recall the definition of hypercohomology groups with values in a sheaf of a differentiable stack, recall the locally constant sheaf, define hypercohomology with values in the constant sheaf as our object of interest
- Show how for representable stacks, hypercohomology corresponds to the cohomology of the sheaf on the manifold
- Define the de Rham cohomology of a differentiable stack and sketch why it agrees with our object of interest
- Sketch why we can compute the de Rham cohomology as the Bott-Shulman-Stasheff complex of an atlas
- Source recommendations: Sections 3.1-3.3 of [1]
- Recommended prerequisites: Know a little about sheaf cohomology on topological spaces, know what Čech cohomology is, know what a spectral sequence is

References

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